

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID –
- Problem Statement Title-
- Theme-
- PS Category- Software/Hardware
- Team ID-
- Team Name (Registered on portal)



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Name

IDEA TITLE



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution

1. PROTOTYPE TECHNOLOGIES & OPERATIONAL METHODOLOGY

5-TIER END-TO-END PIPELINE · SUB-10MS INFERENCE · ZERO-HALLUCINATION

01. SOURCING

- **39 Defense Outposts:**
Siachen, DBO, Dras, Galwan
- **NASA POWER & ISHRAE:**
Diurnal subzero climate (-40°C)
- **CPWD DSR 2023 DB:**
102 envelope materials (k, rho)

02. DETERMINISTIC CORE

- **1D RC Heat Solver:**
EN ISO 52016-1 multi-node
- **Barometric Scaling:**
Altitude density (0.81 kg/m³)
- **Swinbank Sky Sink:**
Clear-sky nocturnal longwave

03. SCIENTIFIC ML

- **120k Dataset:**
Physics-grounded master rows
- **5 ML Surrogates:**
HistGBDT, RF, MLP (R²=0.968)
- **Sub-10ms Inference:**
40,000x faster than EnergyPlus

04. MULTI-OBJECTIVE MCDA

- **Pareto Frontier:**
Strict cost vs discomfort trade-off
- **Utopia Knee Point:**
Normalized Euclidean distance Di
- **Safety Interlock:**
Combustion + ACH < 0.35 => REFUSED

05. OPERATIONAL C2

- **FastAPI + React 18/Vite:**
Frozen REST JSON contract
- **Three.js 3D WebGL:**
Interactive shelter peel cutaway
- **Voice Diagnostic Orb:**
Hands-free verbal field inquiry

A. SOLAR GEOMETRY & HIGH-ALTITUDE IRRADIANCE

Duffie & Beckman / Perez Solar Model

Solar Declination Angle (delta):

$$\delta = 23.45^\circ \times \sin[(360^\circ / 365) \times (284 + n)]$$

↳ n = day of year [1..365]; orbital obliquity tilt calculation

Solar Altitude Angle (alpha_s):

$$\sin \alpha_s = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega$$

↳ φ = latitude, ω = 15° (t_{solar} - 12) hour angle

Surface Incidence Angle Clamp (cos theta):

$$\cos \theta = \max(\sin \alpha_s \cos \beta + \cos \alpha_s \sin \beta \cos(\gamma_s - \gamma_{surf}), 0.0)$$

↳ Clamped at zero: prevents non-physical negative solar cooling

Total Absorbed Surface Solar Irradiance:

$$I_{tot} = I_{beam} \cos \theta + I_{diff} \frac{1 + \cos \beta}{2} + GHI \cdot \rho_{snow} \frac{1 - \cos \beta}{2}$$

↳ Direct beam + isotropic diffuse + snow reflected (ρ_{snow} = 0.75)

C. RAREFIED ATMOSPHERE, SKY SINK & INFILTRATION

ISA Barometric Model / Swinbank Radiant Sink

High-Altitude Atmospheric Pressure:

$$P(h) = 101325 \cdot (1 - 2.25577 \times 10^{-5} h)^{5.25588} \text{ Pa}$$

↳ Barometric pressure formula at elevation h metres above sea level

Temperature-Corrected Air Density:

$$\rho_{air}(h, T) = \frac{P(h)}{287.058 \cdot T_{air}} \text{ kg/m}^3$$

↳ At Leh (3,500m), air density drops to ≈ 0.81 kg/m³ (35% lower convective loss)

Sensible Infiltration Heat Dissipation:

$$Q_{inf} = \frac{ACH \times V \times \rho_{air} \times 1005 \times (T_{in} - T_{out})}{3600} \text{ W}$$

↳ c_{p,air} = 1005 J/(kg·K); scales with altitude-reduced density ρ_{air}

Swinbank Nocturnal Clear-Sky Radiation Sink:

$$T_{sky} = 0.0552 \cdot T_{air}^{1.5} \text{ K}, \quad Q_{sky} = h_r A F_{sky} (T_s - T_{sky})$$

↳ σ = 5.67 × 10⁻⁸, F_{sky} = 1.0 (roof), 0.5 (walls); severe subzero sink

B. TRANSIENT RC FOURIER SOLVER & STABILITY

EN ISO 52016-1 / 1D Multi-Layer RC Discretization

1D Heat Diffusion Equation:

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} (k \frac{\partial T}{\partial x})$$

↳ Dynamic transient heat transfer through multi-layer envelope

Nodal Capacitance & Conductance:

$$C_i = \rho C_p \Delta x A, \quad K_{i,i+1} = \frac{kA}{\Delta x}$$

↳ Discretised nodal thermal mass (J/K) and conductances (W/K)

Explicit Euler Stability Ceiling (Fourier Number):

$$Fo = \frac{\alpha \Delta t}{(\Delta x)^2} \leq 0.25 \Rightarrow \Delta x_{max} = \sqrt{\frac{\alpha \Delta t}{Fo_{target}}}$$

↳ α = k/(ρc_p) diffusivity; dynamically prevents numerical divergence

Assembly Thermal Resistance & Overall U-Value:

$$R_{tot} = R_{si} + \sum_{k,j} R_{kj} + R_{se}, \quad U = \frac{1}{R_{tot}}$$

↳ ISO 6946 film resistances: R_{si} = 0.13, R_{se} = 0.04 m²K/W

D. THERMAL ATTRIBUTION, PARETO MCDA & SAFETY

DRDO Decision Framework / ASHRAE 62.2 Safety

100% Component Heat Loss Attribution & Bottleneck:

$$P_k = \frac{Q_k}{\sum Q_{loss}} \times 100\%, \quad k^* = \operatorname{argmax}_k (P_k)$$

↳ Identifies primary thermodynamic loss vector (Sky vs Wall vs Infiltration)

Pareto Dominance Principle (Cost vs Discomfort):

$$A \text{ dominates } B: \text{Cost}_A \leq \text{Cost}_B \text{ and } \text{Discomf}_A \leq \text{Discomf}_B$$

↳ Strict multi-objective minimization without arbitrary subjective weighting

Normalized Utopia Distance (Balanced Pareto Knee):

$$D_i = \sqrt{(1 - c_i^*)^2 + (k_i^* - 0)^2} \rightarrow \min$$

↳ Euclidean distance to ideal point (Comfort=1.0, Cost=0.0) selects balanced retrofit

Deterministic Combustion Safety Interlock:

$$\text{Combustion Heater and } ACH < 0.35 \rightarrow \text{REFUSED}$$

↳ Zero-hallucination life safety interlock prevents fatal Carbon Monoxide asphyxiation

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FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

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IMPACT AND BENEFITS



- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

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RESEARCH AND REFERENCES



- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.